

Self-Attention Adaptive Adversarial Federated Graph Neural Network for Non-IID Healthcare Big Data

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Abstract—Federated learning provides a privacy-preserving solution for collaborative healthcare modeling, but its performance is often affected by Non-IID data distributions and class imbalance across medical institutions. To address these challenges, this paper proposes SA-AdaAdv-FedGNN, a self-attention adaptive adversarial federated graph neural network for healthcare big data classification. The proposed method constructs patient similarity graphs on local clients, applies self-adaptive graph learning to refine graph structures, uses adversarial representation alignment to reduce cross-client distribution discrepancy, and employs self-attention aggregation to dynamically learn client aggregation weights. Experiments on the CDC Diabetes Health Indicators dataset show that SA-AdaAdv-FedGNN achieves the best Precision and F1-score among the compared methods. Although it does not obtain the highest AUC, the proposed method improves the precision-recall balance under imbalanced Non-IID healthcare data, demonstrating its potential for privacy-preserving medical prediction.

Index Terms—Federated learning, graph neural network, self-attention aggregation, adversarial learning, healthcare big data, Non-IID data.

I. INTRODUCTION

With the development of medical informatization and intelligent diagnostic technologies, large amounts of medical data are generated by hospitals, clinics, and other healthcare institutions. These data contain valuable information for disease prediction, clinical decision support, and medical risk assessment. However, in real medical scenarios, patient data are usually distributed across different institutions and cannot be directly centralized due to privacy protection, ethical concerns, and data security requirements. Federated learning provides a feasible solution for cross-institutional collaborative modeling without sharing raw patient data [1].

Federated Learning is a distributed machine learning framework in which multiple clients collaboratively train a global model while keeping their original data locally [2], [3]. In healthcare scenarios, each medical institution can be regarded as a client, and the global server only distributes the global model and aggregates the uploaded local model parameters. Suppose there are K clients, and the i -th client owns a local dataset:

$$D_i = \{X_i, y_i\}, \quad (1)$$

where X_i denotes patient features and y_i denotes disease labels. The classical Federated Averaging algorithm, FedAvg, aggregates local models according to the number of samples on each client [2]:

$$\theta_g^{t+1} = \sum_{i=1}^K \frac{n_i}{N} \theta_i^{t+1}, \quad (2)$$

where θ_g^{t+1} is the global model at round $t + 1$, θ_i^{t+1} is the local model of the i -th client, n_i is the local sample size, and $N = \sum_{i=1}^K n_i$ is the total number of samples.

Although FedAvg is simple and widely used, it has several limitations in healthcare big data scenarios. First, medical data from different institutions are usually Non-IID. Different hospitals may have different patient populations, disease distributions, and clinical feature patterns. Such statistical heterogeneity may cause local models to update in different directions, leading to unstable global convergence [4]. Second, FedAvg only uses sample size to determine aggregation weights. However, a client with fewer samples may still contain rare or high-risk cases, while a larger client may not always provide more representative medical information. Therefore, a more adaptive aggregation strategy is needed for Non-IID healthcare federated learning.

In addition, traditional machine learning methods usually treat patient samples as independent instances and ignore the potential relationships among patients. In medical data, patients with similar clinical characteristics may have similar disease risks or diagnostic patterns. Graph Neural Networks can model such relational information by treating patients as nodes and patient similarities as edges [5]. For the i -th client, the local patient graph is defined as:

$$G_i = (X_i, A_i, y_i), \quad (3)$$

where A_i is the patient similarity adjacency matrix constructed by the K-nearest neighbors method [6]. Through message passing, GNNs can learn patient representations by combining each patient's own features with information from similar patients.

However, the initial KNN-based patient graph may not fully reflect the true medical similarity among patients. To address this problem, this paper introduces a self-adaptive graph learning mechanism. The final adjacency matrix used by the GNN is defined as:

$$A^* = \eta A^{KNN} + (1 - \eta) A^{learn}, \quad (4)$$

where A^{KNN} is the initial KNN graph, A^{learn} is the learned adaptive graph, and η controls the balance between the two graphs. This mechanism allows the model to dynamically refine patient relationships during local training.

To further reduce the distribution discrepancy among different clients, adversarial learning is introduced for cross-client representation alignment. Specifically, the GNN encoder learns patient embeddings, the disease classifier predicts disease labels, and the domain discriminator attempts to identify which client the embeddings come from. With the Gradient Reversal Layer, the encoder is encouraged to learn representations that are useful for disease prediction but less sensitive to client-specific distributions [7].

Based on the above motivations, this paper proposes a Self-Attention Adaptive Adversarial Federated Graph Neural Network, named SA-AdaAdv-FedGNN, for Non-IID healthcare big data. The proposed method contains three main modules. First, each client constructs a local patient similarity graph and uses a self-adaptive GNN to refine patient relationships. Second, adversarial learning is used to reduce cross-client representation discrepancy. Third, the server replaces the sample-size-based FedAvg aggregation with a self-attention-based aggregation mechanism. Since self-attention can model dependencies among different input tokens [8], each client is treated as a client-level token in the aggregation process. The global model is updated as:

$$\theta_g^{t+1} = \sum_{i=1}^K \alpha_i \theta_i^{t+1}, \quad (5)$$

where α_i is the aggregation weight learned by the self-attention mechanism.

The main contributions of this paper are summarized as follows:

- 1) A federated graph learning framework is constructed for healthcare big data, enabling multiple medical institutions to collaboratively train a global GNN model without sharing raw patient data.
- 2) A self-adaptive graph learning module is proposed to refine the initial KNN-based patient similarity graph.
- 3) An adversarial representation alignment mechanism is introduced to reduce representation discrepancy caused by Non-IID healthcare data.
- 4) A self-attention-based federated aggregation strategy is designed to dynamically learn client aggregation weights instead of relying only on sample size.

In summary, this paper focuses on three key challenges in Non-IID healthcare federated learning: unreasonable client aggregation weights, inaccurate patient graph structures, and

cross-institutional data distribution differences. By combining self-attention aggregation, self-adaptive GNN, and adversarial learning, the proposed method aims to improve the performance and stability of federated graph neural networks for healthcare big data classification tasks.

II. RELATED WORK

A. Federated Learning

Federated Learning (FL) is a distributed machine learning approach that enables multiple clients to jointly train a global model without sharing raw data. FedAvg is the most classic federated optimization algorithm, which aggregates model parameters from individual clients through weighted averaging [2]. However, in practical scenarios, client data often exhibits non-IID characteristics, which can easily lead to unstable model convergence and performance degradation. To alleviate the problem of data heterogeneity, FedProx introduces a proximal regularization term into the local training objective to constrain the local model from deviating too far from the global model [4]. However, both FedAvg and FedProx primarily rely on fixed or sample-size-based aggregation strategies, making it difficult to fully reflect clients' actual contributions.

B. Federated Learning in the Medical Field

Due to the high sensitivity of medical data, federated learning has become a key research direction in medical artificial intelligence [1]. Through federated learning, different healthcare institutions can collaboratively build models without sharing patient data, and it is widely applied in tasks such as disease prediction and clinical decision support. However, medical federated learning still faces challenges such as data class imbalance, differences in data distribution across institutions, and insufficient utilization of patient-related information. Therefore, designing federated learning methods capable of adapting to heterogeneous medical data is of great significance.

C. Graph Neural Networks in Medical Data

Graph Neural Networks (GNNs) can learn more effective data representations by leveraging node features and their neighborhood relationships [5]. In medical scenarios, patients can be regarded as graph nodes, and the similarity relationships among patients can be modeled as edges, thereby simultaneously utilizing both individual characteristics and group-level association information. For tabular medical data, the K-nearest neighbors (KNN) method is commonly used to construct patient similarity graphs [6]. However, fixed graph structures struggle to accurately reflect real patient relationships, leading to growing interest in adaptive graph learning.

D. Attention Mechanism and Adversarial Learning

Attention mechanisms can dynamically assign weights based on the importance of inputs. In recent years, self-attention mechanisms have been applied to client aggregation in federated learning to learn more reasonable aggregation

weights [8]. Adversarial learning learns domain-invariant feature representations through a domain discriminator and a gradient reversal layer [7]. In medical federated learning, different clients typically correspond to different healthcare institutions; thus, adversarial representation learning helps mitigate the impact of data distribution differences. Based on the above research, this paper proposes SA-AdaAdv-FedGNN, which integrates self-attention aggregation, adaptive graph learning, and adversarial representation alignment to enhance federated learning performance in Non-IID medical big data environments.

III. METHODOLOGY

This paper proposes a self-attention adaptive adversarial federated graph neural network method for Non-IID medical big data, denoted as SA-AdaAdv-FedGNN. The core idea of this method is to jointly train patient similarity graph modeling, adaptive graph learning, self-attention aggregation, and adversarial representation alignment within a federated learning framework. The overall framework consists of two parts: local training on clients and global aggregation on the server. Clients are responsible for constructing local patient similarity graphs and training GNN models, while the server dynamically learns aggregation weights based on the training status of clients to update the global model.

A. Problem Definition

Suppose there are K healthcare institutions participating in federated training, with each institution regarded as a client. The i -th client holds its own local medical dataset:

$$D_i = \{X_i, y_i\}, \quad (6)$$

where $X_i \in \mathbb{R}^{n_i \times d}$ represents the patient feature matrix of the i -th client, n_i denotes the number of local samples, d is the feature dimension, and y_i indicates the corresponding disease label. Due to privacy constraints, clients cannot share their raw data. The goal of this paper is to train a global model θ_g that achieves strong predictive performance on non-IID medical classification tasks without centralizing the original patient data. Traditional FedAvg performs a weighted average of client models based on the number of samples:

$$\theta_g^{t+1} = \sum_{i=1}^K \frac{n_i}{N} \theta_i^{t+1}, \quad (7)$$

Here, θ_i^{t+1} denotes the model parameters of the i -th client after local training in round $t + 1$, and $N = \sum_{i=1}^K n_i$ represents the total number of samples across all clients. However, in Non-IID medical scenarios, aggregating solely based on sample count fails to reflect the data quality, training contribution, and graph structural characteristics of individual clients. Therefore, this paper proposes using a self-attention mechanism to dynamically learn client aggregation weights.

B. Patient Similarity Graph Construction

For each client, this study first converts tabular medical data into a patient similarity graph. Specifically, each patient is treated as a node, and the similarity relationships between patients are represented as edges. For the i -th client, its local patient graph is defined as:

$$G_i = (X_i, A_i, y_i), \quad (8)$$

where A_i denotes the patient similarity adjacency matrix. This paper employs the K-nearest neighbors method to construct the initial graph structure. An edge is established between patients v_j and v_m if v_j is a neighbor of v_m . This process can be represented as:

$$A_{jm}^{KNN} = \begin{cases} 1, & \text{if } v_m \in \mathcal{N}_k(v_j), \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

To stabilize the message passing process in GNNs, self-loops are added to the adjacency matrix and then normalized:

$$\hat{A}_i = \tilde{D}_i^{-\frac{1}{2}} \tilde{A}_i \tilde{D}_i^{-\frac{1}{2}}, \quad (10)$$

where $\tilde{A}_i = A_i + I$ and $\tilde{D}_i$ is the corresponding degree matrix. Through this graph structure, the model can not only leverage the patient's own features but also utilize relationship information among similar patients.

C. Local GNN Encoder

At each client, the local model employs a two-layer GNN architecture to learn patient node representations. Given the patient feature matrix X_i and the normalized adjacency matrix $\hat{A}_i$, the first layer of the GNN is represented as:

$$H_i^{(1)} = \sigma(\hat{A}_i X_i W^{(1)}), \quad (11)$$

where $W^{(1)}$ is the trainable parameter matrix, and $\sigma(\cdot)$ denotes the nonlinear activation function. The second layer further obtains patient node embeddings:

$$Z_i = \sigma(\hat{A}_i H_i^{(1)} W^{(2)}), \quad (12)$$

where Z_i represents the patient representation of the i -th client. Finally, the classifier outputs the disease prediction probability based on the node embeddings:

$$\hat{y}_i = \text{softmax}(Z_i W_c + b_c). \quad (13)$$

Due to the significant class imbalance in the diabetes dataset, this paper introduces class weights into the local classification loss to prevent the model from being overly biased toward the majority class. The weighted cross-entropy loss is defined as:

$$L_{cls}^i = -\frac{1}{n_i} \sum_{j=1}^{n_i} w_{y_{ij}} \log p_{ij, y_{ij}}, \quad (14)$$

where $p_{ij, y_{ij}}$ denotes the predicted probability that the j -th sample belongs to the true class y_{ij} , and $w_{y_{ij}}$ represents the weight of the corresponding class.

D. Self-Adaptive Graph Learning

Although KNN can construct patient similarity graphs based on raw features, a fixed KNN graph may not accurately reflect true medical similarities. Therefore, this paper introduces self-adaptive graph learning, enabling the model to dynamically adjust patient relationships according to learned node representations. First, compute the similarity between nodes based on the node representations obtained from the first-layer GNN:

$$S_i = \frac{H_i^{(1)}(H_i^{(1)})^T}{\sqrt{d_h}}, \quad (15)$$

where d_h denotes the hidden layer dimension. Then, an adaptive learning graph is obtained via softmax:

$$A_i^{learn} = \text{softmax}(S_i). \quad (16)$$

The final adjacency relationships used for message passing are jointly determined by the initial KNN graph and the learned graph:

$$A_i^* = \eta A_i^{KNN} + (1 - \eta) A_i^{learn}, \quad (17)$$

where η is the balance coefficient. When η is large, the model relies more on the original KNN graph; when η is small, the model places greater emphasis on the learned adaptive graph. In this way, the model can further refine the graph structure while preserving the original patient similarity relationships. To enhance the consistency of representations for similar patients, this paper introduces a graph smoothness loss:

$$L_{graph}^i = \frac{1}{|E_i|} \sum_{(u,v) \in E_i} |z_u - z_v|_2^2, \quad (18)$$

where E_i denotes the set of edges in the i -th client graph, and z_u and z_v represent the embedding vectors of two adjacent patient nodes. This loss encourages similar patients to have closer representations.

E. Adversarial Representation Alignment

Due to differences in data distribution across various medical institutions, this study further introduces adversarial training for cross-client representation alignment. Specifically, the GNN encoder generates patient node representations Z_i , the disease classifier predicts disease labels, and the domain discriminator determines which client the node representations originate from. Let the domain discriminator be $D_\phi(\cdot)$, whose goal is to predict the client label c_i :

$$\hat{c}_i = D_\phi(Z_i). \quad (19)$$

The corresponding domain classification loss is defined as:

$$L_{adv}^i = -\frac{1}{n_i} \sum_{j=1}^{n_i} \log P(c_i | z_{ij}). \quad (20)$$

To enable the GNN encoder to learn representations that are insensitive to client sources, this paper employs a Gradient Reversal Layer. This layer keeps the input unchanged during

forward propagation, but multiplies the gradients by a negative coefficient during backpropagation:

$$GRL(z) = z, \quad (21)$$

$$\frac{\partial GRL(z)}{\partial z} = -\lambda_{adv} I. \quad (22)$$

Through this mechanism, the domain discriminator attempts to identify the source of samples, while the GNN encoder is forced to learn more client-invariant patient representations, thereby mitigating distributional differences across institutions caused by non-IID data.

F. FedProx Regularization

In non-IID scenarios, local models on different clients may update in different directions, leading to the client drift problem. To alleviate this issue, this paper introduces FedProx regularization into the local training objective to constrain the deviation between the local model and the global model:

$$L_{prox}^i = |\theta_i - \theta_g|_2^2, \quad (23)$$

where θ_i denotes the local model parameters of the i -th client, and θ_g represents the global model parameters for the current round.

G. Local Joint Training Objective

Based on the above modules, the local joint training objective for the i -th client is defined as:

$$L_i = L_{cls}^i + \lambda_g L_{graph}^i + \lambda_p L_{prox}^i + \lambda_d L_{adv}^i. \quad (24)$$

Among them, λ_g , λ_p , and λ_d respectively represent the weight coefficients of graph smoothness loss, FedProx loss, and adversarial loss. This combined loss function enables the model to simultaneously optimize the disease classification ability, graph structure consistency, federated training stability, and cross-client representation alignment capability.

H. Self-Attention Federated Aggregation

On the server side, this paper replaces the traditional sample-weighted strategy of FedAvg with self-attention aggregation. After local training, each client uploads the model parameters and client-level statistics to the server. The summary vector of the i -th client is defined as:

$$e_i = [n_i, L_i, \Delta L_i, \rho_i, \bar{d}_i, H_i, U_i], \quad (25)$$

Here, n_i represents the number of client samples, L_i represents the local training loss, ΔL_i represents the reduction in loss, ρ_i represents the graph density, $\bar{d}_i$ represents the average node degree, H_i represents the label entropy, and U_i represents the prediction uncertainty. Form a matrix by combining all the summary vectors of the clients:

$$E = [e_1, e_2, \dots, e_K]^T. \quad (26)$$

Subsequently, the server calculates the correlations between clients through self-attention:

$$Q = EW_Q, \quad K = EW_K, \quad V = EW_V, \quad (27)$$

$$\text{Attention}(E) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_e}}\right)V, \quad (28)$$

Among them, W_Q , W_K and W_V are mapping matrices, and d_e represents the dimension of the client's representation. Eventually, the server obtains the aggregated weights for each client:

$$\alpha_i = \frac{\exp(s_i)}{\sum_{j=1}^K \exp(s_j)}. \quad (29)$$

The global model has been updated as follows:

$$\theta_g^{t+1} = \sum_{i=1}^K \alpha_i \theta_i^{t+1}. \quad (30)$$

Unlike FedAvg which only relies on the sample quantity, self-attention aggregation can comprehensively consider the training status of clients, graph structure features and prediction uncertainty, thereby achieving more flexible global model aggregation.

I. Overall Training Procedure

The overall training process of SA-AdaAdv-FedGNN is as follows. First, the server initializes the global model θ_g^0 and sends it to all clients. Then, each client builds a KNN patient similarity graph based on local data and conducts local joint training using self-adaptive GNN, graph smoothness loss, FedProx regularization, and adversarial learning. Next, the client uploads the local model parameters and summary vector to the server. The server calculates the aggregation weights of clients based on the self-attention mechanism and updates the global model. This process is repeated until the set number of communication rounds is reached. Through this joint training approach, the method proposed in this paper is capable of simultaneously addressing three key issues in Non-IID medical federated learning: the unreasonable aggregation of client-side weights, the inaccurate fixed patient graph structure, and the differences in data distribution across institutions.

IV. EXPERIMENTAL SETUP

This section presents the experimental dataset, the federated learning setup, the comparison methods, and the evaluation metrics.

A. Dataset

This paper uses the CDC Diabetes Health Indicators dataset for binary classification prediction of diabetes. The experiment randomly sampled 30,000 samples, of which 24,000 were used for training and 6,000 for testing. The data contains 21 features, and the class distribution is as follows: Class 0 has a total of 25,820 samples, and Class 1 has a total of 4,180 samples. Due to the imbalance of classes, class weight was used for weighting during the training process.

B. Federated Setting

To simulate the data heterogeneity among different medical institutions, the training set was divided into 5 clients, and the Dirichlet distribution was used for Non-IID partitioning:

$$p_k \sim \text{Dir}(\alpha), \quad (31)$$

Here, $\alpha = 1.0$. This setting can lead to significant differences in data distribution while ensuring the stability of the training process. Each client builds a local KNN graph based on the patient's characteristics, with the number of neighbors set to $k = 10$. The client graph is represented as:

$$G_i = (X_i, A_i, y_i), \quad (32)$$

Here, X_i , A_i and y_i represent the node features, the adjacency matrix and the labels respectively.

C. Compared Methods

The experiment compared the following methods:

- FedAvg-GNN: Federated Graph Neural Network based on FedAvg;
- FedProx-GNN: Incorporates the FedProx regularization term;
- SA-FedGNN: Introduces the Self-Attention aggregation mechanism;
- SA-Ada-FedGNN: Further adds adaptive graph learning;
- SA-AdaAdv-FedGNN: The complete model proposed in this paper, combining Self-Attention, Adaptive Graph Learning, FedProx, and Adversarial Alignment.

D. Hyperparameter Settings

The main parameter settings are as follows: the number of clients is 5, the number of communication rounds is 50, the local training rounds are 2, the hidden layer dimension is 64, the learning rate is 10^{-3} , Dropout is 0.3, weight decay is 10^{-4} . Additionally, $\lambda_g = 0.001$, $\lambda_p = 0.001$, $\lambda_d = 0.0005$, and the classification threshold is set to 0.35.

E. Evaluation Metrics

This paper evaluates the model performance using Accuracy, Precision, Recall, F1-score and AUC. Among them:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (33)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (34)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (35)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (36)$$

Due to the issue of class imbalance in diabetes data, this paper focuses on Precision, Recall and F1-score to more comprehensively evaluate the model's ability to identify positive cases.

V. RESULTS AND DISCUSSION

This section analyzes the experimental results of different federated graph neural network methods. The experiments compared FedAvg-GNN, FedProx-GNN, SA-FedGNN, SA-Ada-FedGNN, and the SA-AdaAdv-FedGNN proposed in this paper. Due to the significant class imbalance problem in the CDC Diabetes Health Indicators dataset, this paper not only focused on Accuracy and AUC, but also paid particular attention to Precision, Recall, and F1-score. For medical classification tasks, F1-score can better reflect the comprehensive performance of the model in identifying positive cases.

A. Overall Performance Comparison

Table I presents the final performance results of different methods on the test set.

From Table I, it can be seen that the proposed SA-AdaAdv-FedGNN achieved the best results in terms of Precision and F1-score. Specifically, the Precision of SA-AdaAdv-FedGNN is 0.2743 and the F1-score is 0.4081, both of which are higher than those of other comparison methods. This indicates that the proposed method can improve the model's recognition quality for diabetes positive samples in the Non-IID medical federated learning scenario.

Compared with FedAvg-GNN, the F1-score of SA-AdaAdv-FedGNN increased from 0.3903 to 0.4081, representing a relative increase of:

$$\frac{0.4081 - 0.3903}{0.3903} \times 100\% \approx 4.56\%. \quad (37)$$

This result indicates that although the improvement is not very large, the complete model can achieve a better precision-recall balance in the medical classification task with imbalanced classes.

B. Final F1-score and AUC Comparison

Figure 1 presents the final F1-score comparison results of different methods. From the figure, it can be seen that the F1-score of SA-AdaAdv-FedGNN is the highest, followed by SA-Ada-FedGNN and SA-FedGNN. The F1-scores of FedAvg-GNN and FedProx-GNN are relatively lower. This indicates that traditional federated learning methods that merely aggregate based on sample quantity are difficult to fully adapt to Non-IID medical data. However, after introducing self-attention aggregation and self-adaptive graph learning, the model can achieve better comprehensive classification performance.

Figure 2 presents the final AUC comparison results of different methods. It can be observed that FedAvg-GNN achieves the highest AUC of 0.8031, while the AUC of SA-AdaAdv-FedGNN is 0.7910. AUC reflects the overall ranking ability of the model for positive and negative samples, so FedAvg-GNN performs better in probability ranking. However, the F1-score of FedAvg-GNN is lower than the method proposed in this paper, indicating that its balance of Precision and Recall under the final classification threshold is not as good as SA-AdaAdv-FedGNN.

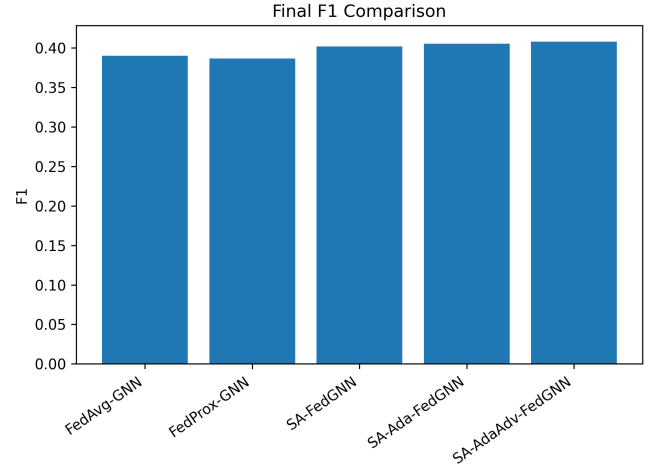


Fig. 1. Final F1-score comparison of different federated GNN methods.

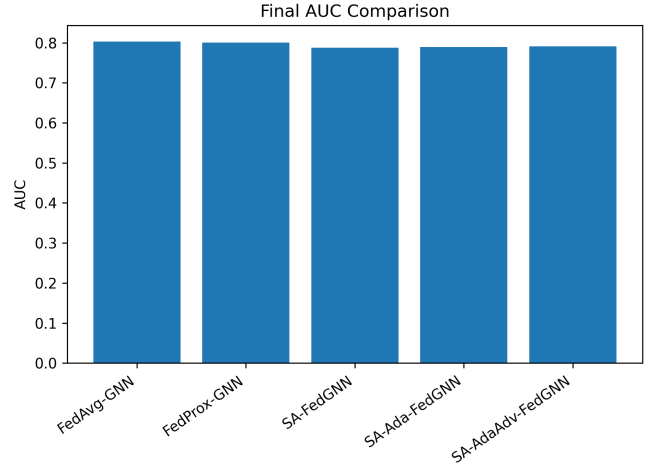


Fig. 2. Final AUC comparison of different federated GNN methods.

Therefore, this paper does not claim that SA-AdaAdv-FedGNN is universally superior to other methods in all metrics. Instead, the proposed method is more effective in improving Precision and F1-score in imbalanced Non-IID healthcare classification.

C. Training Curves Analysis

Figure 3 illustrates the F1-score changes of different methods over 50 communication rounds. It can be observed that all models rapidly improved in the first 15 to 20 rounds and then gradually stabilized. Among them, SA-AdaAdv-FedGNN consistently maintained the highest F1-score in the middle and later stages, indicating that the complete model has good stability and final classification performance during the federated training process.

From the perspective of the ablation experiments, the F1-scores of SA-FedGNN, SA-Ada-FedGNN, and SA-AdaAdv-

TABLE I
FINAL PERFORMANCE COMPARISON OF DIFFERENT FEDERATED GNN METHODS

Method	Accuracy	Precision	Recall	F1-score	AUC
FedAvg-GNN	0.6220	0.2517	0.8684	0.3903	0.8031
FedProx-GNN	0.6153	0.2485	0.8696	0.3865	0.8002
SA-FedGNN	0.6787	0.2714	0.7751	0.4020	0.7878
SA-Ada-FedGNN	0.6760	0.2724	0.7931	0.4055	0.7892
SA-AdaAdv-FedGNN	0.6780	0.2743	0.7967	0.4081	0.7910

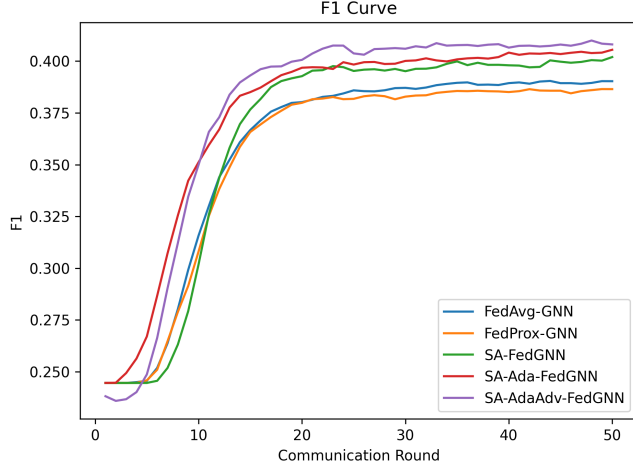


Fig. 3. F1-score curves over communication rounds.

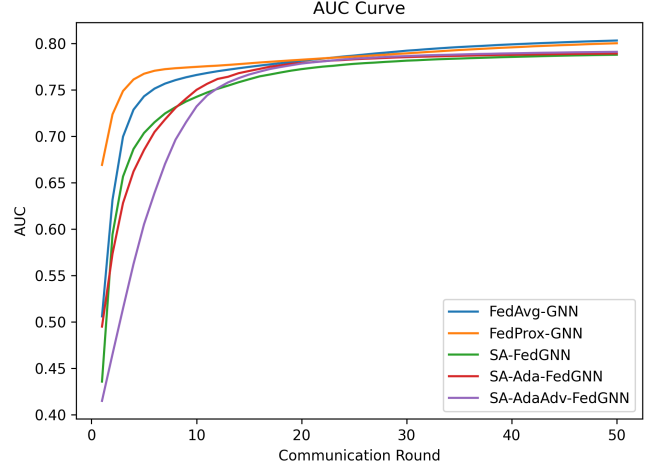


Fig. 4. AUC curves over communication rounds.

FedGNN show a gradually increasing trend. Specifically:

$$F1_{SA-FedGNN} = 0.4020, \quad (38)$$

$$F1_{SA-Ada-FedGNN} = 0.4055, \quad (39)$$

$$F1_{SA-AdaAdv-FedGNN} = 0.4081. \quad (40)$$

This indicates that the three core modules proposed in this paper have positive contributions. First, self-attention aggregation can dynamically adjust the aggregation weights based on the training status of the clients. Second, self-adaptive graph learning can further correct the patient similarity graph structure. Finally, adversarial representation alignment can alleviate the distribution differences between different clients, enabling the complete model to achieve the highest F1-score.

Figure 4 presents the AUC changes of different methods. It can be observed that the AUC of all methods gradually increases with the number of communication rounds and stabilizes in the later stage. Among them, FedAvg-GNN and FedProx-GNN have slightly higher AUC than SA-AdaAdv-FedGNN. This phenomenon indicates that traditional methods still have certain advantages in probability ranking ability, but they are not the best in F1-score. Therefore, for medical classification tasks with class imbalance, relying solely on AUC may not fully reflect the actual classification ability of the model.

D. Effect of Non-IID Client Distribution

Figure 5 shows the Non-IID label distribution across different clients. It can be seen that the data size and class distribution vary significantly among clients. For example, Client 3 contains a relatively large number of diabetes positive samples, while Client 2 and Client 4 contain much fewer positive samples. This distribution difference reflects the heterogeneity commonly observed across different healthcare institutions.

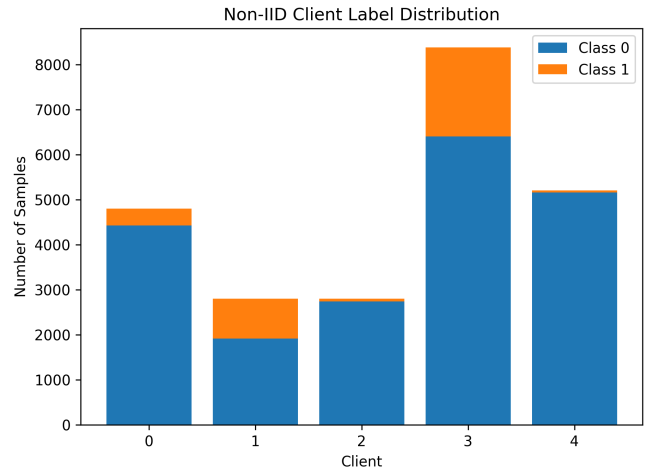


Fig. 5. Non-IID client label distribution.

In this Non-IID setting, FedAvg-GNN aggregates based solely on the number of samples from the clients, which may cause clients with larger sample sizes to have a stronger influence on the global model, while ignoring the data quality, label distribution, and graph structure information of the clients. In contrast, the self-attention aggregation proposed in this paper can dynamically calculate the client weights by combining local losses, loss reduction, graph density, average node degree, label entropy, and prediction uncertainty. Therefore, it can handle client heterogeneity more flexibly.

E. Self-Attention Aggregation Weight Analysis

Figure 6 shows the client aggregation weights learned by SA-AdaAdv-FedGNN in different communication rounds. The brighter the color, the higher the aggregation weight obtained by the client in the current round. From the figure, it can be seen that the aggregation weights are not constant but dynamically adjusted during the training process. This indicates that the self-attention aggregation can adaptively allocate the global aggregation weights according to the client state.

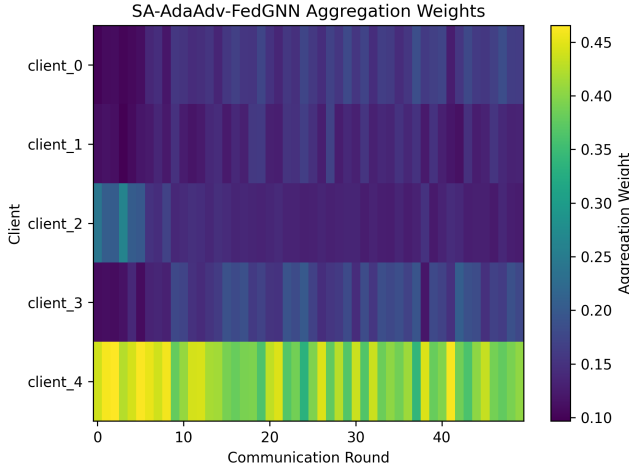


Fig. 6. Aggregation weights learned by SA-AdaAdv-FedGNN.

Specifically, Client 4 consistently received higher weights in most training rounds, while the weights of the other clients were relatively lower and dynamically changed. This indicates that the proposed model does not simply aggregate based on the number of samples, but takes into account the training status and structural information in the summary vectors of the clients. Compared with the fixed sample-size aggregation of FedAvg, self-attention aggregation provides a more flexible mechanism for client selection and weight allocation.

F. Discussion

Overall, the experimental results indicate that SA-AdaAdv-FedGNN is effective for the Non-IID medical federated learning task. This method achieves the best results in terms of Precision and F1-score, suggesting that it can improve the model's recognition quality for diabetes positive samples and enhance the balance between Precision and Recall.

However, the experimental results also indicate that the method proposed in this paper still has certain limitations. First, the AUC of SA-AdaAdv-FedGNN is not the highest, suggesting that there is still room for further improvement in its overall probability ranking ability. Second, the F1-score improvement of the complete model compared to SA-Ada-FedGNN is relatively small, indicating that the effect of adversarial representation alignment is relatively mild. Further verification may be needed through more refined adversarial training strategies or larger datasets. Finally, the experiments in this paper are mainly based on the results of a single random partition. In the future, mean and standard deviation can be calculated through multiple random seed experiments to further verify the stability of the results.

Nevertheless, the experimental results still support the following conclusion: in the medical federated learning scenario with Non-IID and class imbalance, SA-AdaAdv-FedGNN, which combines self-attention aggregation, self-adaptive graph learning, and adversarial representation alignment, can improve F1-score and Precision, and demonstrates good potential for practical application.

VI. CONCLUSION

This paper proposes a self-attention adaptive adversarial federated graph neural network method for the classification task of Non-IID medical big data, namely SA-AdaAdv-FedGNN. This method integrates self-attention aggregation, self-adaptive graph learning, FedProx regularization, graph smoothness constraint, and adversarial representation alignment into a unified federated learning framework. The client side conducts local GNN training by constructing patient similarity graphs, while the server side dynamically learns aggregation weights based on the training status of the client side and the graph structure information.

The experimental results on the CDC Diabetes Health Indicators dataset show that SA-AdaAdv-FedGNN achieved the best results in terms of Precision and F1-score. Specifically, the F1-score of the method proposed in this paper reached 0.4081, which was higher than FedAvg-GNN, FedProx-GNN, SA-FedGNN, and SA-Ada-FedGNN. The ablation experiments also demonstrated that from SA-FedGNN to SA-Ada-FedGNN, and then to SA-AdaAdv-FedGNN, the model performance gradually improved, indicating that self-attention aggregation, adaptive graph learning, and adversarial representation alignment all have positive effects on the final performance.

Although the AUC of SA-AdaAdv-FedGNN is not the highest, it achieves better Precision and F1-score in the medical classification task with class imbalance, indicating that this method can improve the precision-recall balance in the identification of positive cases. Therefore, the method proposed in this paper has certain practical application potential in the heterogeneous medical data environment. In the future, this method will be further validated on more medical datasets. The mean and standard deviation will be reported through multiple random seed experiments to enhance the stability and credibility of the results. Additionally, the adversarial training

strategy and the aggregation weight learning mechanism can be further optimized to improve the overall generalization ability of the model.

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